Sample size calculation ImPRESS study v2

Sample size calculation

26th of March 2019

Introduction

This document provides details of the sample size calculation for the Imaging Perfusion Restrictions from Extracellular Solid Stress (ImPRESS), an open-label losartan study. The caluclations are based on simulations.

This updated version is after response from SLV.

Study design

The suggested study design is an individual stepped-wedge, randomized, assessor-blinded, dose-finding trial on three indications. Study A is divided into recurrent glioblastoma and newly diagnosed glioblastoma patients, while Study B includes patients with brain metastases.

Following discussions the following treatment regime is suggested:

Study A (recurrent glioblastoma)

Group	Baseline	Days 1 to 14	Days 15 to 28	Days 29 to 42	Days 43 to 180
Group 1					
Group 2					
Group 3					

Note:
Red is no treatment, blue is treatment with losartan

Note that all treatment groups are treated with either 25 mg/day, 50 mg/day or 100 mg/day losartan in a 1:1:1 alocation ratio. We denote the treatment groups as Group 1: 25 mg/day, Group 1: 50 mg/day etc. There are 9 treatment groups in total.

Study A (newly diagnosed glioblastoma)

Group	Baseline	Days 1 to 14	Days 15 to 28	Days 29 to 42	Days 43 to 180
Group 1					
Group 2					
Group 3					

Note:
Red is no treatment, blue is treatment with losartan

Note that all treatment groups are treated with either 25 mg/day, 50 mg/day or 100 mg/day losartan in a 1:1:1 alocation ratio. We denote the treatment groups as Group 1: 25 mg/day, Group 1: 50 mg/day etc. There are 9 treatment groups in total.

Study B (Brain metastases)

Group	Baseline	Days 1 to 90	Days 91 to 180	Days 181 to 270	Days 271 to 360
Group 1					
Group 2					
Group 3					

Note:
Red is no treatment, blue is treatment with losartan

All patients will be treated with 50 mg/day losartan. There are 3 treatment groups in total.

Model

The underlying model is defined as

$$\Delta RPI_{ij} = A_i + \mu_{trt} + \mu_j + \epsilon_{ij}$$

where ΔRPI_{ij} is the change from baseline Restricted Perfusion Imaging value, measured as a percentage of the optimal value (-100% to 100%) of patient $i = 1..N$ at timepoint j, A_i is the random intercept of patient i, μ_{trt} is the treatment effect , μ_j is the expected RPI at timepoint j, and ϵ_{ij} is the residual. Note that we assume the treatment effect is equal between the treatment doses, such that the μ_j denotes the average treatment effect over the three doses.

Assumptions and model parameters

Every sample size calculation needs assumptions on the effect size and the variance of the observations.

Effect size

We assume the effect size to be 20%. We also assume that the effect size is equal between the treatment doses in Study A. However, we want to assess the power of different effect sizes, and assess the effect sizes of 5%, 10% and 15% in addition.

Variance

The variance is difficult to assess, but we assume an overall standard deviation of 15%, split with a within and between patient standard deviaiton of 10.6%

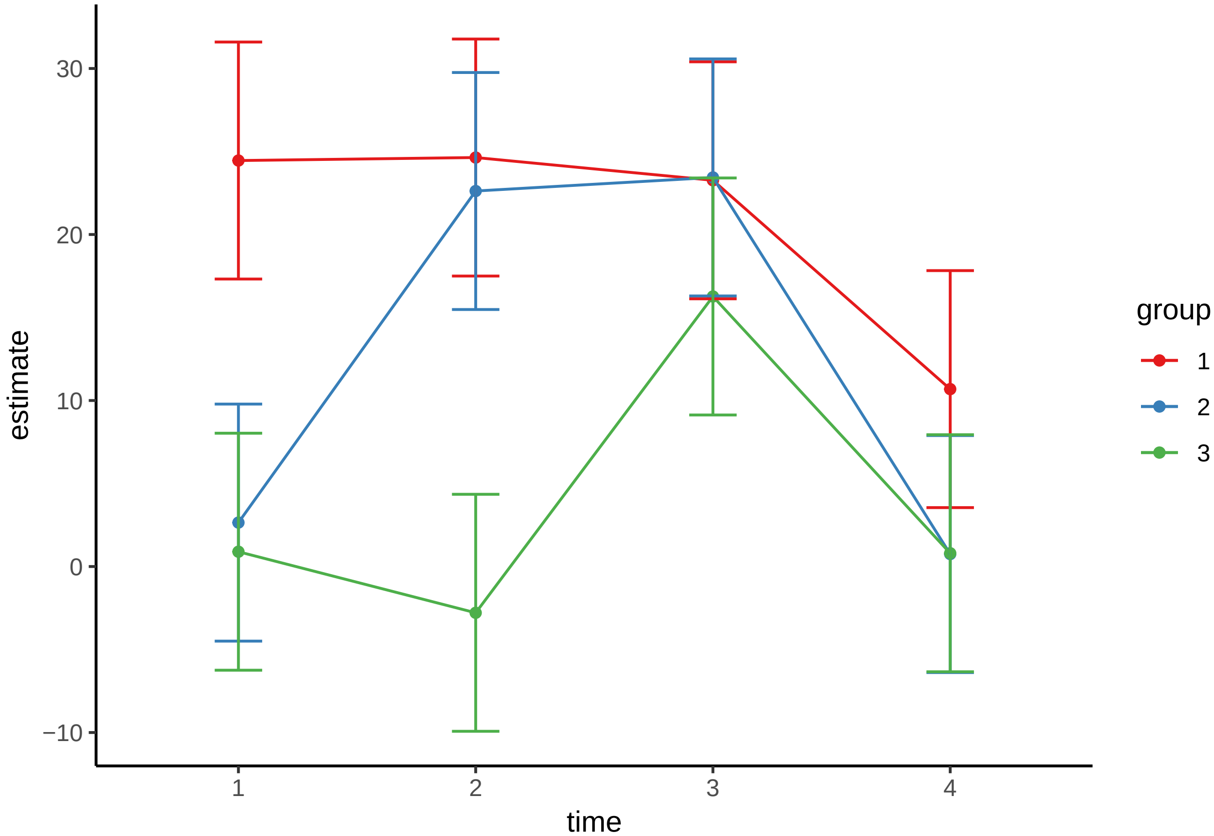
Simulations

We run simulations with different sample sizes using this design and these assumptions. We run 1000 simulated trials for each sample size. For each simulation we estimate the parameters of interest (treatment effect) and the p-value from a mixed model as defined previously. The power is calculated as the proportion of trials with significant treatment effect. For each simulated trial we simulate 6, 12 or 15 patients in each intervention group. As previously noted, patients in each treatment group in Study A receives randomly one of three doses. In the calculations, the doses are pooled to assess the overall treatment effect of losartan.

Results

Result of one simulated trial

Below is a plot of one randomly simulated trial with 18 in each group, design is as for Study A (recurrent glioblastoma) and treatment effect is 20%.



Power calculation

Based on the simulations, we form tables to estimate the power.

Sample size per group	Effect size	Total sample size	Power	Estimate
6	5	18	0.226	5.154
6	10	18	0.615	9.905
6	15	18	0.937	15.091
6	20	18	0.994	20.062
12	5	36	0.390	5.178
12	10	36	0.890	9.885
12	15	36	0.999	14.923
12	20	36	1.000	19.958
15	5	45	0.437	5.047
15	10	45	0.961	10.070
15	15	45	1.000	15.007
15	20	45	1.000	19.843
18	5	54	0.504	4.944
18	10	54	0.984	10.002
18	15	54	1.000	15.045
18	20	54	1.000	19.943

Conclusion

Following the simulations, we see that with an effect size of 20% we have very high power for all sample sizes. With 12 in each group, we see that we gain a sufficient power even when the effect size is as low as 10%.

References